

Robin Erb

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EDUCATION

University of Copenhagen (KU)
Semester Abroad (M.Sc. Computer Science)

Feb 2026 – July 2026
Copenhagen, Denmark

Heidelberg University
Master of Science in Data and Computer Science | GPA: 1.3
– Specialization: Machine Learning

Apr 2025 – May 2027
Heidelberg, Germany

University of Konstanz
Bachelor of Science in Computer Science | GPA: 1.4
– Specialization: Data Science and Human-Computer Interaction
– **Thesis (Grade: 1.0):** Developed a hybrid input device for 2D/3D interaction (Spatial Mouse )
– Conducted a comprehensive user study and analytical evaluation using Python

Oct 2021 – Mar 2025
Konstanz, Germany

Jacob-Grimm-Schule Kassel
A Levels (Abitur) | GPA: 1.8

Sep 2017 – Jun 2020
Kassel, Germany

PROFESSIONAL EXPERIENCE

Campus Five
Freelance Software Developer

Sep 2025 – Present
Remote

- **Project: Automated Middleware and ERP Integration**
 - Built real-time middleware to sync sales, returns, and inventory between local ERP and Shopify
 - Automated catalog management including product drafting, publishing, and archiving processes
 - Engineered end-to-end system architecture using Google Cloud Functions and Firestore database
- **Project: AI-Driven Sales Automation**
 - Implemented OCR data extraction for vehicle documents to support the automated sales pipeline
 - Assisted in deploying an AI chatbot for inquiries and built a staff verification dashboard

University of Konstanz (HCI Department)
Student Assistant

Sep 2023 – Mar 2025
Konstanz, Germany

- Developed the "Spatial Mouse" prototype, including 3D CAD modeling
- Engineered Arduino-based hardware in C, involving circuit soldering and component assembly
- Built a dedicated virtual test environment for the prototype in Unity

Steinbock
Bouldering Instructor (Part-time)

Sep 2021 – Jul 2023
Konstanz, Germany

- Taught climbing techniques and safety protocols in youth bouldering classes

Kletterwerk Radolfzell
Voluntary Service (Federal Volunteer Service - BFD)

Sep 2020 – Aug 2021
Radolfzell, Germany

- Instructed school classes in bouldering and rope climbing

PUBLICATIONS

S. Hubenschmid, J. Zagermann, **R. Erb**, et al. "SpatialMouse: A Hybrid Pointing Device for Seamless Interaction Across 2D and 3D Spaces." In *Proc. of the 31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*. DOI: 10.1145/3756884.3766047

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, Java, C++, SQL, R
Cloud & Tools: Google Cloud Functions, Firebase/SQL, Git, Docker, Unity, CAD
ML & Automation: OCR, AI-Bots, PyTorch, Scikit-learn, Pandas
Languages: German (Native), English (Fluent), French (Basic)
Interests: Rock Climbing, Volleyball, Snowboarding, DJing